SilkReverie

AI virtual try-on and authentic xiangyunsha apparel, delivered as a downloadable Android app

Business Plan

Mobile Application Development · Course Assignment, Phase 1

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Team	Zhong Haiyang · Yao Daben · Ding Siqi · Wen Juyue
Institution	South China Normal University, Aberdeen Institute of Data Science and Artificial Intelligence
Stage	Working build: full purchase loop and try-on demonstrated; Android app in build
Partner	Wanxiang Youyi, a xiangyunsha menswear brand

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1. Executive summary

SilkReverie is an Android app that lets a shopper see how a piece of xiangyunsha clothing looks on her own body before she buys it, and gives every garment a readable product code so that authenticity can be checked without a database. It is aimed at the gap between a craft that sells well offline and a buying experience online that cannot show what the fabric actually does.

The plan starts narrowly on purpose. We work with one menswear brand, Wanxiang Youyi, from the producing region; we list a small, standard-sized menswear range; and we put three things in one flow that are currently sold separately: a try-on image, a size choice that takes seconds, and a code that proves the piece is real.

- **The problem.** Counterfeits dilute the brand, static photos cannot convey the lustre and drape of the gauze, and young shoppers have no low-risk way to try the look.
- **The product. An Android app with four modules: virtual try-on, purchase flow, product codes and fabric notes. The try-on prompts are written for this fabric rather than borrowed from a generic fashion model.**
- **The model.** Commission on resale (5% of an average order of about RMB 800), plus paid digital services and content production for the merchant, plus small-volume merchandise.
- **What we must prove next.** That a try-on image on this fabric is good enough to change a purchase decision, and that generating it costs less than the commission on one order.

A working build (v0.11) already demonstrates the full loop — browse, size selection, order and try-on; the Android app continues on the same cloud backend and interaction code. Every figure in this document comes either from the partner's price list or from an assumption that is labeled as such and will be tested in the pilot.

2. The problem

2.1 A craft that is hard to photograph

Xiangyunsha is a silk dyeing and finishing craft that dates from the Song dynasty. It is dyed with gambier (shuliang) and coated with river mud, in more than thirty steps, without chemical dye. It is a national-level intangible cultural heritage, and it looks different from ordinary silk: a directional metallic sheen, a starched hand-feel, a print that sits in the weave.

2.2 Three obstacles in online retail

Counterfeits. Imitations are sold alongside genuine pieces, and the ordinary buyer has no quick way to tell them apart, so the brand's credibility is diluted by products it did not make.

The fabric resists the format. Lustre and drape do not survive a flat product photo. The gap between the listing and what arrives is large, and returns are correspondingly high.

No entry point for young buyers. Promotion is one-dimensional and mostly offline. A younger customer has neither a reason to look nor a way to try the look without visiting a store.

2.3 Who pays for the gap

The merchant absorbs it as returns and discount pressure on a high-priced item. The shopper absorbs it as risk on an RMB 800 purchase. The craft absorbs it as a narrowing group of wearers. A design that only helps one of the three would not survive contact with the shop's economics.

3. The solution

3.1 Four modules in one flow

Each module answers one of the obstacles above. They are deliberately not separate apps or separate journeys.

- **Virtual try-on.** The shopper uploads a half-body or full-body photo and sees the garment on herself. The generation prompt is adjusted for this fabric's texture, print and sheen, rather than using a generic template.
- **Purchase flow.** Browse, size selection, bag, order confirmation and address management sit in one sequence, so the shopper does not lose the item while comparing.
- **Product codes.** Every garment carries a readable code; style, color, size and year can be read by eye and summarized by the shop without a database query.
- **Fabric notes.** The product page states material, dyeing method and producing area. Short craft videos and live streams from the drying yard are planned for a later iteration.

3.2 Why the try-on has to be built for this fabric

A general fashion try-on model produces plausible clothes, not this fabric: the silhouette drifts and the sheen becomes ordinary silk. The design therefore spends its effort on the prompt, not only on the model choice.

1. **Upload a photo.** A half-body or full-body photo of the shopper, replaceable at any time.
2. **Choose the garment.** The category selects a prompt template automatically.
3. **Generate.** A cloud call to a generation model; the plan is to use Kling 3.0 Omni.

4. **Archive the result.** The image is stored per user, grouped by try-on task, and can be revisited.

Five constraints carry most of the quality, and all of them are written into the prompt: four scene templates (full body, upper body, lower body, outerwear); a fixed silhouette, garment length and sleeve length so the model cannot redesign the piece; an explicit description of gambier dyeing, jacquard texture and directional sheen; side lighting so the weave is visible; and per-user isolation of stored results.

3.3 The product code

A single string replaces four lookups. The example below is the coding scheme we will use; the same rule generates order numbers, membership numbers and try-on task numbers.

`YY-CS-M-26-001-BK-L`

YY brand · CS category (shirt) · M menswear · 2026 · style 001 · BK black · L size

For the shop, the code is read without a query: style, color, size and year are legible, and sales can be summarized along any of those dimensions. For the platform, the same discipline applies to orders (OD-date-source-serial), memberships (MB-year-serial) and try-on tasks (FT-date-serial). Serial numbers are allocated in one place so that concurrent orders cannot collide.

3.4 Scope: menswear first

Menswear has fewer styles, more standard sizing and a simpler range than womenswear. That makes it the cheapest place to find out whether a try-on image on this fabric is convincing, and whether the code scheme holds when it meets a real product list. Womenswear follows only if those two questions are answered.

4. Market analysis

4.1 The industry is moving toward higher value

Xiangyunsha production is shifting toward higher added value and a wider range of products, and the space for integrating production, research and sales is opening up. In the producing regions the base is already larger than a craft-scale operation: in Wengyuan, Guangdong, workshop programs built with paired assistance support point to more than 1 million meters shipped in 2024 and an annual output value above RMB 20 million, according to regional reporting compiled by the team (see Sources). At the same time, AI try-on and intelligent shopping assistants have become a recognized direction in e-commerce, which is what makes a try-on-first storefront possible for a small team.

4.2 Policy support

Three national documents describe the same direction: digital use of cultural resources, new cultural experience settings, and support for traditional industries that combine with new technology.

- **Cultural digitization.** Opinions on Implementing the National Cultural Digitization Strategy, issued by the CPC Central Committee General Office and the State Council General Office in 2022.
- **Heritage protection.** The 14th Five-Year Plan for Intangible Cultural Heritage Protection (Ministry of Culture and Tourism) encourages digital technology to extend heritage transmission and sales channels.
- **Digital cultural industry.** Guidance on Promoting the High-Quality Development of the Digital Cultural Industry supports new digital formats and the combination of traditional culture with AI.
- **Local base.** Guangdong promotes the creative transformation of Lingnan culture, and Foshan Shunde has an established xiangyunsha industrial base and heritage protection system.

4.3 Who the customers are

Three groups, each with a different reason for hesitating at checkout.

Segment	Profile	Hesitation, and our answer
Core growth segment	New Chinese-style wearers; 25–35; tier-1 and tier-2 cities; mostly women	Worries that the piece will look dated on her and that sizing is guesswork. Answer: try-on first, size selection in a half-screen sheet.
Higher-ticket buyers	Quality menswear buyers; 30–45; business and gifting occasions	Worries about counterfeits at a high price. Answer: one-code-per-item authenticity and a menswear range that can be inspected.
Existing loyal customers	Long-standing xiangyunsha wearers; 45+; loyal to origin and craft	Buys offline; needs a simple, trustworthy online reorder path, often placed by a son or daughter. Answer: an app that opens instantly, with craft content that continues the offline relationship.

Reference price: the partner's menswear shirts are listed at RMB 759–899, about RMB 800 on average.

Persona. Lin Wei, 28, a UI designer in Guangzhou, wears new Chinese-style outfits to weekend exhibitions and posts on Xiaohongshu. She saves xiangyunsha pieces to her wishlist but never orders: at RMB 800 she cannot judge the drape from photos, and she has returned "silk" before that turned out to be imitation. SilkReverie is built for her — an AR-free try-on image made from her own photo, a product code she can read, and a checkout that takes three taps.

4.4 How the existing channels compare

Dimension	Marketplaces	Heritage content accounts	Offline showrooms	SilkReverie (app)
Craft explanation	Absent	Strong	Strong	Built into product pages
Virtual try-on	No fabric fit	None	None	Tuned for xiangyunsha
Transaction	Complete	Redirect, drop-off	Limited by location	Completes in the app
Authenticity	Mixed with imitations	None	Manual	Product code

No existing channel combines craft explanation, try-on and verified transaction in one place. That combination, not the try-on alone, is the position we are claiming.

4.5 What this analysis does and does not support

The public data support the supply side well: the producing base is large, policy supports digital channels, and try-on is an accepted e-commerce direction. They do not provide a credible number for how many shoppers want to try this fabric online, and we will not quote a market size we cannot defend. Instead the first market is measured from the partner: one menswear range, an average order value of about RMB 800, and a test of whether try-on lifts conversion among shoppers who have already opened a product page. The pilot target is 100 completed orders, which is small enough to be achievable and large enough to show whether the flow works.

5. Business model

5.1 Revenue lines

Line	What it covers	Pricing
Apparel resale	Xiangyunsha garments sold through the app (average order about RMB 800)	5% commission on order value
Digital services	Shopper profiling, sentiment monitoring, try-on service for the merchant	RMB 20,000–80,000 per year
Merchandise	Derivative products at about RMB 100 each	Retail margin
Content production	Documentary shooting and live-stream operations	RMB 20,000–50,000 per project

Terms as agreed with the partner brand. Service and content prices are the bands we quoted to the partner; neither is contracted yet.

5.2 Unit economics of one order

One RMB 800 order produces RMB 40 of commission (5%) and leaves the merchant RMB 760, which is why the model can be offered without an upfront fee. The question that decides viability is the cost of producing one try-on image: if a generation plus storage and bandwidth costs less than RMB 40, an order that is converted by try-on pays for its own generation. Below that threshold the model scales with orders; above it, try-on has to be limited to high-intent shoppers or a paid add-on.

5.3 Cost structure

- **Usage-based and small at pilot volume: cloud function calls, database reads and image storage on a serverless cloud backend, plus model calls per try-on.**
- **Time-based:** content production, merchant onboarding and support, all currently done by the four founders.
- **Deliberately absent: advertising spend, warehouse, inventory, and server operations. The app holds no stock; the merchant ships.**

5.4 What has to be true

1. **Try-on quality.** The generated image is close enough to the real garment that a shopper uses it to decide, not to browse.
2. **Cost per generation.** Below the RMB 40 commission on an average order.
3. **Merchant economics.** Zero upfront cost and a 95% share of the order value, so the shop has no reason to refuse a trial.
4. **Repeat use.** Shoppers return to see a second garment, which turns a one-off try-on into a browsing habit.

6. Go-to-market plan

The four channels are sequenced rather than run in parallel, because each one depends on material the previous one produces.

1. Content first. Craft, styling and dyeing-yard material published continuously on Xiaohongshu and Douyin, pointing to the app. This is the only channel that creates demand before the app is opened.

2. Try-on images as distribution. A generated image is something a shopper is willing to post. Shared to WeChat Moments and group chats, it does the work that paid placement would otherwise do, at no cost per impression.

3. Documentary and live stream. Filmed with the merchant in the drying yard and the workshop. It builds the authenticity the code alone cannot convey, and it is also a paid service we sell to the same merchant.

4. Campus and private groups. University communities, heritage societies and offline markets supply the first cohort of users who can be watched closely while the flow is still rough.

No advertising budget is planned. Product images and try-on material come from the partner; content is produced in-house; the first audience is reached through campus and heritage communities. What we measure in the pilot is narrow and explicit: try-on completion rate, orders per 100 product-page views, and return rate against the merchant's current online baseline.

7. Technology and delivery plan

7.1 Stack

Layer	Choice
Client	Cross-platform Android app built with a uni-app style framework; ships with goods, order and try-on modules
Backend	Serverless cloud backend: cloud functions for goods, orders, users, user data and try-on; managed database and storage
Generation	Kling 3.0 Omni, called from a cloud function
Delivery	Git repository; versioning and release automated on merge to main (planned)

A serverless cloud backend is chosen for practical reasons rather than novelty: no server to rent, no filing, no operations work. An interface layer lets the pages switch between demo data and the real cloud functions, so design and implementation can proceed in parallel.

7.2 Build order

1. **Frame, sub-packages, design system.** Address management and weak-network handling come with the frame, not later.
2. **Goods list and detail.** Half-screen size selection and the bag.
3. **Order confirmation.** Order number generation and the server-side price recalculation.
4. **Try-on link.** Photo upload, cloud generation, archived result.
5. **Code scheme.** Wired through goods, orders and try-on records.

7.3 Data protection

Face photographs are sensitive personal information, so the design calls for four rules: images are stored per user, consent is requested explicitly before upload, results are retained only for a stated period, and the model API key lives in the cloud function environment rather than in the repository. Order amounts are recalculated on the server, so a price passed by the client is ignored.

7.4 Current status

A working build already implements the interfaces, flows, code scheme and architecture; the Android app packaging starts this phase on the same cloud backend. The first technical checkpoint is a try-on test on ten partner garments, before any interface work, because a weak result there would change the product.

8. Financial plan (illustrative)

The figures below are a scenario built on stated assumptions, not a forecast. They exist to show the shape of the business and the size of the first step, and will be replaced by actuals after the pilot.

	Year 1	Year 2	Year 3
Partner stores	1	3	6
Paid orders	600	3,000	8,000
Average order value	RMB 800	RMB 800	RMB 800
Resale commission (5%)	RMB 24,000	RMB 120,000	RMB 320,000
Content projects	1	2	3
Content revenue	RMB 20,000	RMB 50,000	RMB 75,000
Digital service contracts	0	1	3
Digital service revenue	RMB 0	RMB 40,000	RMB 150,000
Merchandise units (RMB 100 each)	300	1,500	3,000
Merchandise margin (25%)	RMB 7,500	RMB 37,500	RMB 75,000
Total revenue	RMB 51,500	RMB 247,500	RMB 620,000

Assumptions: orders per store rise as try-on is proven (about 50 per month in year 1 rising to about 110 in year 3); content and service prices sit inside the bands quoted to the merchant; merchandise carries a 25% margin on RMB 100 items.

Costs are not modelled line by line, because two inputs are unknown until the pilot: the cost of one model call and actual cloud usage. Both are usage-based and small at these volumes. The threshold that matters is stated in section 5.4: cost per generation must stay below RMB 40. Fixed costs are near zero, so any commission revenue covers operations; the risk is not the break-even point but the quality of the try-on image.

No external funding is sought at this stage. The team covers development and content production with its own time, and the merchant carries inventory and shipping. If the pilot works, the first use of money would be model credits and shooting costs, both recoverable from the first service contract.

9. Risks and mitigations

Risk	Why it matters	Mitigation
Try-on fidelity	A wrong silhouette or a generic sheen makes the feature less useful than an ordinary product photo.	Ten-garment test before interface work; per-category prompt templates; side lighting instruction; regenerate option; the image is labeled as a simulation.
Generation cost	If one try-on costs more than RMB 40, each converted order loses money.	Measure cost per generation in the pilot; cap free generations per shopper; cache results per user and garment.
Merchant cooperation	The shop must supply images, sizes and samples, and it moves slowly.	Zero upfront cost and revenue share; start with one range; written milestones so the shop knows what it gets and when.
Authenticity claims	A code is only trustworthy if it is tied to a real product record.	Codes are generated in one place and stored with the garment record; the page says what the code proves and what it does not.
Personal data	Face photos are sensitive and easy to mishandle.	Explicit consent, per-user isolation, retention limit, keys outside the code, review before any public launch.
Team capacity	Four students, part-time, with coursework in parallel.	Narrow scope (menswear only), fixed build order, small batches of work, and a pilot target of 100 orders rather than a full launch.

10. Roadmap

Phase	Timing	Deliverables	Decision gate
Try-on test	First month	Generations on ten partner garments; cost per image measured	Image convincing and cost below RMB 40; otherwise revise the design before building
Build	This semester	Frame and sub-packages; goods list and detail; size selection; bag; order confirmation	A complete purchase can be made on a real device
Pilot	After build	Try-on link live for one menswear range; 100 completed orders; return rate tracked	Try-on lifts orders per 100 product views, or the feature is repositioned
Expand	2027	Three to five merchant brands; fit recommendation; content services	Unit economics hold across more than one merchant
Replicate	2028	Additional heritage apparel categories; overseas Chinese market research	A second category repeats the first without redesign

11. Team

Four students from the Aberdeen Institute of Data Science and Artificial Intelligence at South China Normal University: Zhong Haiyang, Yao Daben, Ding Siqi and Wen Juyue. The work is divided into four streams that meet weekly — product definition and the merchant relationship, engineering, content and visual design, and documentation and research.

The team's advantage is not size but adjacency: the people who write the try-on prompt are the people who talk to the merchant about returns, which is why the feature is specified around a business problem rather than a model demo. The constraint is time; the plan is scoped to a single menswear range and a 100-order pilot for that reason.

12. Appendix: definitions, assumptions and sources

Definitions

Xiangyunsha. Gambiered Canton gauze; a silk dyed with gambier and coated with river mud, in more than thirty steps, without chemical dye.

Try-on. A generated image placing a specific garment on a shopper's own photograph.

Product code. The readable identifier described in section 3.3.

Assumptions to be tested

- Average order value of about RMB 800, from the partner's listed prices (RMB 759–899).
- Orders per store rising from about 50 to about 110 per month as try-on is proven.
- Merchandise margin of 25% on an average RMB 100 product.
- Cost per try-on generation below RMB 40; to be measured, not assumed in any launch decision.

Sources

- Opinions on Implementing the National Cultural Digitization Strategy, CPC Central Committee General Office and State Council General Office, 2022.
- 14th Five-Year Plan for Intangible Cultural Heritage Protection, Ministry of Culture and Tourism.

- Guidance on Promoting the High-Quality Development of the Digital Cultural Industry, Ministry of Culture and Tourism.
- Guangdong provincial programs on the creative transformation of Lingnan culture; Foshan Shunde xiangyunsha industrial base (public reporting).
- Wengyuan county xiangyunsha output figures for 2024 (over 1 million meters shipped, over RMB 20 million output value), compiled by the team from public reporting.
- Partner product listings and quoted service prices, Wanxiang Youyi, 2026.